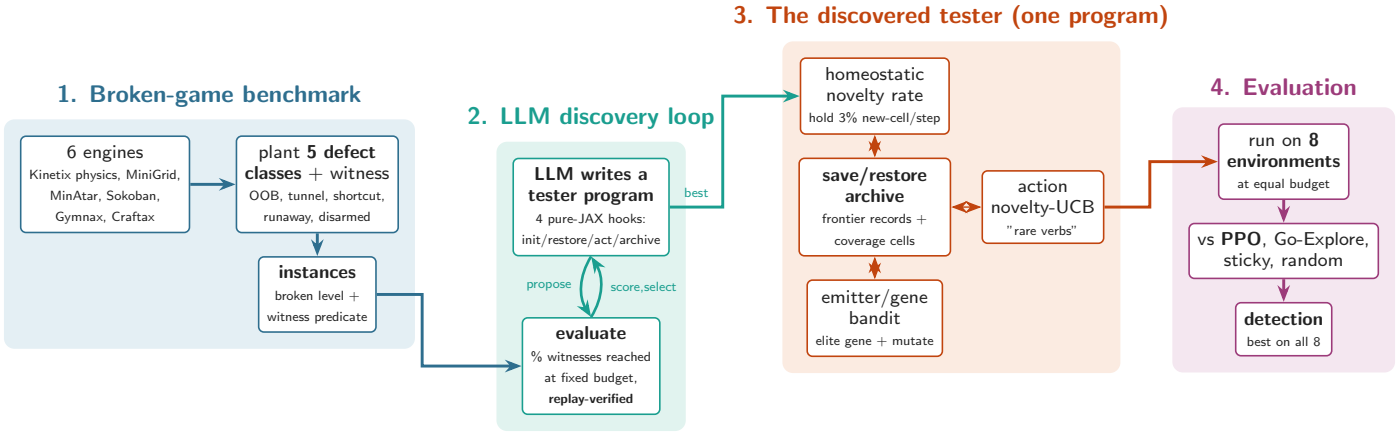
1. Broken-game benchmark
6 engines
Kinetix physics, MiniGrid, MinAtar, Sokoban, Gymnax, Craftax
plant 5 defect classes + witness
OOB, tunnel, shortcut, runaway, disarmed
instances
broken level + witness predicate
2. LLM discovery loop
LLM writes a tester program
4 pure-JAX hooks: init/restore/act/archive
best
propose
score,select
evaluate
% witnesses reached at fixed budget, replay-verified
3. The discovered tester (one program)
homeostatic novelty rate
hold 3% new-cell/step
save/restore archive
frontier records + coverage cells
action novelty-UCB
"rare verbs"
emitter/gene bandit
elite gene + mutate
4. Evaluation
run on 8 environments
at equal budget
vs PPO, Go-Explore, sticky, random
detection
best on all 8